

VINAYAK BALASUBRAMANIAN

Chennai, India · +91 99621 02414 · vinayakbala32@gmail.com · vinayakb.netlify.app
Cybersecurity Professional · GRC & Compliance Automation · SOC Operations · Secure Software Development

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student at VIT Chennai and Cybersecurity professional with hands-on experience spanning Security Operations, Governance, Risk & Compliance (GRC), and full-stack security software development. Proven track record architecting compliance automation pipelines, tuning enterprise SIEM/SOAR platforms to reduce false positives, and co-authoring peer-reviewed security research. Incoming Master's student in Cybersecurity at Monash University (2027). Pursuing advanced offensive and defensive security certifications including BTL1.

EDUCATION

Monash University — Master of Cybersecurity

Expected: 2027

Melbourne, Australia · Incoming postgraduate student

Vellore Institute of Technology (VIT) — B.Tech, Computer Science

Jun 2022 – May 2026

Chennai, Tamil Nadu · CGPA: 7.70 / 10.00

- Relevant coursework: Computer Networks, Operating Systems, Cryptography & Network Security, DSA, DAA, Software Engineering
- Active member: CodeChef VIT, CYSCOM (Cybersecurity Community), FYI Chapter

Maharishi Vidya Mandir SSS — HSC & SSC

March 2022

Chennai, Tamil Nadu · Stream: PCM + Computer Science · SSC: 95% (Centum in Mathematics) · HSC: 85%

WORK EXPERIENCE

GRC Intern & Software Developer | Raksha Technologies

Chennai, TN · Jan 2026 – Present

Leading compliance audit participation and simultaneously engineering custom GRC automation software — a rare dual mandate combining security governance and software engineering.

- Contribute directly to SOC 2 audits: control reviews, evidence collection, and control gap analysis against Trust Service Criteria.
- Architect and develop Python-based automation tools for firewall rule reviews and centralized approval workflows, reducing manual effort.
- Building full-stack compliance software that automates and streamlines the end-to-end auditing lifecycle for enterprise clients.
- Translate audit findings into actionable remediation roadmaps aligned with industry compliance frameworks (SOC 2, ISO 27001 principles).

SOC Analyst Intern | Noventiq Valuepoint

Chennai, TN · Jun 2025 – Jul 2025

Embedded within an enterprise Security Operations Centre, performing real-time threat monitoring, log analysis, and SIEM/SOAR platform tuning.

- Analyzed security logs and triaged alerts using IBM QRadar and Securonix SNYPR, correlating events via the MITRE ATT&CK framework to identify and escalate incidents.
- Gained deep operational exposure to flow vs. event data, tenant/domain separation, and reference sets for contextual threat enrichment in QRadar.
- Tuned correlation rules and reference sets to reduce false positives by 20% in repeated-login-based offense categories, improving SOC detection accuracy.
- Worked with FortiSOAR for alert automation and incident orchestration, building foundational SOAR playbook knowledge.

Student Research Assistant | VIT Chennai — under Dr. Ilavendhan A

Chennai, TN · Dec 2024 – Jun 2025

Conducted independent cybersecurity research focused on securing IoT healthcare infrastructure for critical patient monitoring environments.

- Researched cybersecurity vulnerabilities in IoT healthcare ecosystems, with focus on securing sensor data pipelines for comatose patient monitoring systems.
- Designed and implemented encryption algorithms and authorization protocols to prevent unauthorized data access and ensure end-to-end data integrity.

- Co-authored a research paper submitted to IEEE Access — a peer-reviewed, high-impact IEEE journal with global readership.

KEY PROJECTS

Multimodal Brain Tumor Classification — Hybrid Deep Learning & Ensemble Model

- Developed an image classification pipeline using CNN for feature extraction combined with PCA and GLCM texture descriptors.
- Integrated CatBoost and LightGBM ensemble classifiers into a hybrid architecture, achieving mean accuracy of 0.86 across 4 classes.
- Classified MRI scans into Pituitary Tumor, Meningioma, Glioma, and No Tumor — with direct application to clinical diagnostic tooling.

Advanced Threat Detection Lab — Wazuh, Sysmon & MITRE ATT&CK

- Deployed a full Wazuh stack (indexer, manager, dashboard) on a Google Cloud Ubuntu instance integrated with a Windows VM Wazuh agent.
- Instrumented the Windows endpoint with Sysmon to capture system-level telemetry: process creation, network connections, file modifications.
- Simulated adversary TTPs mapped to MITRE ATT&CK, authored custom Wazuh detection rules, and validated end-to-end alert fidelity.

Custom C2 (Command & Control) Framework — In Development

- Designing and building a custom C2 server from first principles to deepen understanding of post-exploitation infrastructure, OPSEC tradecraft, and network covertness techniques.

Healthcare Drug Interaction & Symptom Classifier

- Built an AI-powered classifier that identifies drug interactions, symptoms, and side effects for healthcare decision support.
- Applied NLP and classification techniques to medical datasets, aligning with secure and responsible AI development practices.

TECHNICAL SKILLS

Languages:	Python (Advanced), SQL, Bash / Shell Scripting, JavaScript
SIEM / SOAR:	IBM QRadar, Securonix SNYPR, FortiSOAR — correlation tuning, rule authoring, incident triage
GRC & Compliance:	SOC 2, ISO 27001 principles, control review, evidence collection, gap analysis, audit workflows
Offensive Security:	Ethical Hacking, OSINT, OPSEC, custom C2 infrastructure, TryHackMe Jr. Pentester pathway
Defensive Security:	Wazuh, Sysmon, MITRE ATT&CK framework, custom detection engineering, threat hunting
Software Dev:	Full-stack compliance tooling, Python automation (firewall reviews, approval workflows), REST APIs
Cloud & Infra:	Google Cloud Platform (GCP), Ubuntu server administration, Windows endpoint hardening
AI / ML:	CNN, PCA, GLCM, CatBoost, LightGBM, ensemble methods, NLP classifiers
Core Skills:	Risk analysis, threat intelligence, incident response, technical writing, effective communication

CERTIFICATIONS & TRAINING

- Blue Team Level 1 (BTL1) — Security Blue Team [Currently Pursuing]
- QRadar SIEM Fundamentals — IBM
- Essentials of Ethical Hacking — EC-Council
- TryHackMe: Jr. Penetration Tester Pathway — Completed

RESEARCH & PUBLICATIONS

Co-author — Cybersecurity in IoT Healthcare: Securing Sensor Data for Comatose Patient Monitoring Systems
Submitted to IEEE Access (Peer Review) · Supervised by Dr. Ilavendhan A, VIT Chennai · 2025